

MATH I (Fall 2026)

WEEK 1: INDEPENDENT STUDY ASSIGNMENTS (Homework) ¶

About the Independent Study Assignments

The Independent Study Assignments are homework problems and assignments that will help you learn the week's content more deeply by solving problems and expanding your knowledge and skills. *Your answers will not be submitted and will not be graded.* You may solve the problems individually or collaborate with your fellow students or ask the assistance from an AI tool (to help you clarify the concepts, not providing the solution without you thinking about it first).

Week 1 - Assignments

The following problems/projects cover the material presented in Week 1. Choose to do as many as you wish (the more the better):

Assignment #1. [Optional assignment]

The following is an *optional assignment* – you can engage with it if you are familiar with AI-assistants, or if you want to start learning what AI-assistants can do for you in a context-specific domain. The exercise is as follows:

Interact with an AI-assistant (such as ChatGPT, Claude, DeepSeek, Gemini, etc.), by structured conversation, through the following steps:

a) Think of the most important point of the lecture's content and ask the AI-assistant to evaluate your answer (you may ask the AI-assistant to evaluate it (usually, they do the evaluation in terms of content and clarity, but you can also ask for depth and complexity, or other criteria you may choose). The AI-assistant will not grade your input but it will provide content and perspectives to help you better understand the material.

The following example of a simple prompt shows how you can ask (prompt) the AI-assistant:

Your (simple) prompt: "In the first lecture of Math I course, the most important point is the presentation of the number systems." Can you evaluate this statement of mine?

b) Create your own problem (on the lecture's material) and ask the AI-assistant (of your choice) to evaluate it in terms of difficulty and depth.

c) Solve your proposed problem by yourself. You may discuss your problem with fellow students. d) Ask the AI-assistant for a solution to your problem but check its validity (that is, check if the AI-assistant's solution is correct).

If you find wrong outcomes, discrepancies or contradictory statements, converse with the AI-assistant to mention these points, asking for explanations, to resolve them. If you cannot resolve the wrong outcomes, please consult the instructor.

e) Compare your solution with AI-assistant's solution.

f) [optional] Ask the AI-assistant for an alternative solution and compare it to the original one and to your solution.

A personal note (Prof. G. Neofotistos) about this assignment: This type of assignment was

Assignment #2 : Smallest number set

For each number, identify the smallest set among $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ to which it belongs.

0, -7 , $\frac{5}{8}$, 3.25, $\sqrt{16}$, $\sqrt{7}$, $-\sqrt{9}$, π , $\frac{\pi}{2}$, i , $4 - 3i$.

Assignment #3: True or false?

Determine whether each statement is true or false. Correct every false statement.

1. $\mathbb{N} \subseteq \mathbb{Z}$
 2. $\mathbb{Z} \subseteq \mathbb{Q}$
 3. $\mathbb{R} \subseteq \mathbb{Q}$
 4. $\mathbb{Q} \subseteq \mathbb{R}$
 5. $\mathbb{R} \subseteq \mathbb{C}$
 6. Every integer is a natural number.
 7. Every rational number is real.
 8. Every complex number is real.
 9. Every real number can be written as a complex number.
 10. A number cannot be both rational and real.
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Assignment #4: Set diagram

Draw a nested-set diagram representing

$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$.

Place each of the following numbers in the correct region:

5, -2 , $\frac{2}{3}$, $\sqrt{2}$, π , $-4i$, $3 + 2i$.

Assignment #5: Ordering real numbers

Arrange the following numbers in increasing order:

$$-\sqrt{5}, \quad -2, \quad -\frac{3}{2}, \quad 0, \quad \sqrt{2}, \quad \frac{3}{2}, \quad \frac{\pi}{2}.$$

Justify any comparison that is not immediately obvious.

Assignment #6: Intervals

Write each set using interval notation.

1. $x \in \mathbb{R} : -3 \leq x < 4$
2. $x \in \mathbb{R} : x > 2$
3. $x \in \mathbb{R} : x \leq -1$
4. $x \in \mathbb{R} : -2 < x \leq 5$
5. $x \in \mathbb{R} : x \neq 3$

Then represent each set on the real number line.

Assignment #7: From intervals to inequalities

Write each interval using inequalities.

- a. $[-2, 5)$, b. $(-\infty, 4]$, c. $(1, \infty)$,
d. $(-3, 0] \cup [2, 6)$, e. $\mathbb{R} \setminus [-1, 1]$.
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Assignment #8: Distance between real numbers

The distance between two real numbers a and b is

$$d(a, b) = |a - b|.$$

Calculate:

1. $d(3, 8)$;
2. $d(-2, 5)$;
3. $d(-4, -9)$;
4. $d(\sqrt{2}, \sqrt{8})$.

Then verify numerically that

$$d(a, c) \leq d(a, b) + d(b, c)$$

for $a = -3$, $b = 2$ and $c = 7$.

Assignment #9: Rational or irrational?

Determine whether each number is rational or irrational. Justify your answer.

- a. $\sqrt{49}$, b. $\sqrt{18}$, c. $3 + \sqrt{2}$,
d. $2\sqrt{5}$, e. $\sqrt{2} + \sqrt{8}$, f. $(\sqrt{3})^2$,

g. $(1 + \sqrt{2})(1 - \sqrt{2})$, h. $\frac{1}{\sqrt{2}}$.

Assignment #10: Basic complex arithmetic

Given

$$z_1 = 3 + 2i, \quad z_2 = 1 - 4i,$$

calculate:

$$z_1 + z_2, \quad z_1 - z_2, \quad z_1 z_2, \quad \frac{z_1}{z_2}.$$

Write every answer in the form $a + bi$.

Assignment #11: Powers of i

Evaluate:

$$i^2, \quad i^3, \quad i^4, \quad i^5, \quad i^{18}, \quad i^{27}, \quad i^{100}.$$

Describe the repeating pattern and formulate a method based on division of the exponent by 4.

Assignment #12: Complex conjugate and modulus

For each number, find its conjugate and modulus.

$$3 + 4i, \quad -2 + 5i, \quad 6i, \quad -7.$$

Verify for each example that

$$z\bar{z} = |z|^2.$$

Assignment #13: Complex division

Express each result in algebraic form $a + bi$.

a. $\frac{1}{1+i}$, b. $\frac{3+2i}{2-i}$,
c. $\frac{4-i}{1+3i}$, d. $\frac{5i}{2-i}$.

Assignment #14: Quadratic equations

Solve over $\mathbb{C}$:

a. $x^2 + 9 = 0$, b. $x^2 - 4x + 13 = 0$,
c. $2x^2 + 4x + 10 = 0$, d. $x^4 - 16 = 0$.

Indicate which solutions are real and which are nonreal complex numbers.

Assignment #15: Complex plane

Represent the following numbers in the complex plane:

$$2 + 3i, \quad -1 + 4i, \quad -3 - 2i, \quad 4 - i.$$

For each number, determine:

1. its real part;
 2. its imaginary part;
 3. its modulus;
 4. the quadrant in which it lies.
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Assignment #16: Polar form

Write each complex number in trigonometric form

$$z = r(\cos \theta + i \sin \theta)$$

and exponential form

$$z = re^{i\theta}.$$

1. $1 + i$
2. $-1 + i$
3. $-\sqrt{3} - i$
4. $2i$
5. -4

Give θ in radians.

Assignment #17: de Moivre's formula

Use

$$[r(\cos \theta + i \sin \theta)]^n = r^n (\cos(n\theta) + i \sin(n\theta))$$

to calculate:

1. $(1 + i)^6$;
2. $(\sqrt{3} + i)^4$;
3. $(-1 + \sqrt{3}i)^5$.

Check one result by algebraic multiplication.

Assignment #18: Roots of a complex number

Find all solutions and represent them in the complex plane.

$$\text{a. } z^3 = 1, \quad \text{b. } z^4 = 1, \quad \text{c. } z^3 = -8.$$

Explain why the n -th roots are equally spaced around a circle.

Assignment #19: Write an electric charge in scientific and engineering notations

The magnitude of an electric charge is $2e$ (remember that $e = 0.0000000000000000001602176634\text{ C.}$)

Write the magnitude of this electric charge in a) scientific notation, and b) in engineering notation.

Assignment #20: How many of the SI prefixes do you remember? (fill the "??" in the following Table)

Engineering notation and SI prefixes

Power	Prefix	Symbol
10^{12}	??	T
10^9	giga	G
10^6	??	M
10^3	??	k
10^0	—	—
10^{-3}	milli	m
10^{-6}	??	μ
10^{-9}	??	n
10^{-12}	??	p
10^{-15}	femto	f
10^{-18}	atto	a

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Independent Study Assignments prepared by Prof. G. Neofotistos